Agile Software Engineering and DevOps – Lab 8

Title

Hands-on Lab on Docker and Docker Hub

Lab Objectives

By the end of this lab, students will be able to:

1. Understand **containerization using Docker**
2. Create and run a **Docker container**
3. Write a simple **Dockerfile**
4. Build a **Docker image**
5. Push the image to **Docker Hub**
6. Understand the difference between:
    - Image vs Container
    - Local vs Remote registry

1. Install Docker Desktop

• Download from: https://www.docker.com/products/docker-desktop/ • Install and start Docker

System Requirements

• Windows 10/11 (64-bit) / macOS / Linux
• Minimum 4GB RAM
• Virtualization enabled

2. Create Docker Hub Account

• Sign up at: https://hub.docker.com/

3. Verify Installation

Run in terminal:

```
docker --version
```

docker run hello-world
Expected:

- Docker version displayed
- “Hello from Docker!” message

Lab Execution

1. Run Your First Container

Task 1: Run a Basic Container

docker run hello-world

```
Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent
it
    to your terminal.
```

Task 2: Run an Interactive Container

docker run -it ubuntu bash
Inside container:

echo "Hello from container"
exit

```
C:\Users\SIDDESHWARAPRASAD KR>docker run -it ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
b40150c1c271: Pull complete
92842f25412d: Download complete
Digest: sha256:c4a8d5503dfb2a3eb8ab5f807da5bc69a85730fb49b5cfca2330194ebcc41
c7b
Status: Downloaded newer image for ubuntu:latest
root@87e581e101b0:/# echo "Hello from computer"
Hello from computer
root@87e581e101b0:/# exit
exit
```

Task 3: List Containers

docker ps
docker ps -a

```
C:\Users\SIDDESHWARAPRASAD KR>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\SIDDESHWARAPRASAD KR>docker ps -a
CONTAINER ID   IMAGE         COMMAND    CREATED              STATUS
             PORTS     NAMES
87e581e101b0   ubuntu        "bash"     About a minute ago   Exited (0) 48 s
econds ago             wonderful_pascal
5d8c02f91c02   hello-world   "/hello"   4 minutes ago        Exited (0) 4 mi
nutes ago              practical_perlman
df3730ba7aa4   hello-world   "/hello"   8 minutes ago        Exited (0) 8 mi
nutes ago              vigilant_rhodes
```

2. Create a Simple Application Students create a simple app:

Task 1: Create File

mkdir docker-lab
cd docker-lab
Create app.py:

print("Hello from Docker App")

```
C:\Users\SIDDESHWARAPRASAD KR\Desktop\docker-lab>echo print("Hello from Dock
er App") > app.py

C:\Users\SIDDESHWARAPRASAD KR\Desktop\docker-lab>type app.py
print("Hello from Docker App")
```

3. Create Dockerfile

Task 1: Create Dockerfile

Create a file named:

Dockerfile
Add:

FROM python:3.10-slim
WORKDIR /app
COPY app.py .

CMD ["python", "app.py"]

```
C:\Users\SIDDESHWARAPRASAD KR\Desktop\docker-lab>echo FROM python:3.10-slim
> Dockerfile

C:\Users\SIDDESHWARAPRASAD KR\Desktop\docker-lab>echo WORKDIR /app >> Docker
file

C:\Users\SIDDESHWARAPRASAD KR\Desktop\docker-lab>echo COPY app.py . >> Docke
rfile

C:\Users\SIDDESHWARAPRASAD KR\Desktop\docker-lab>echo CMD ["python", "app.py
"] >> Dockerfile

C:\Users\SIDDESHWARAPRASAD KR\Desktop\docker-lab>type Dockerfile
FROM python:3.10-slim
WORKDIR /app
COPY app.py .
CMD ["python", "app.py"]
```

4. Build and Run Docker Image

Task 1: Build Image

docker build -t docker-lab-app .

```
C:\Users\SIDDESHWARAPRASAD KR\Desktop\docker-lab>docker build -t docker-lab-
app .
[+] Building 11.8s (9/9) FINISHED                      docker:desktop-linux
 => [internal] load build definition from Dockerfile                    0.1s
 => => transferring dockerfile: 119B                                    0.0s
 => [internal] load metadata for docker.io/library/python:3.10-slim     3.2s
 => [auth] library/python:pull token for registry-1.docker.io           0.0s
 => [internal] load .dockerignore                                       0.0s
 => => transferring context: 2B                                         0.0s
 => [1/3] FROM docker.io/library/python:3.10-slim@sha256:26fc00cf5fd6   7.4s
 => => resolve docker.io/library/python:3.10-slim@sha256:26fc00cf5fd6   0.1s
 => => sha256:bed04db721b4ee2f43a7fc743a2fd7ec6181f 13.82MB / 13.82MB   2.3s
 => => sha256:935d60c8480a502328adefa7698ca61261c8dc0 1.29MB / 1.29MB   2.3s
 => => sha256:ddf3b231f5678d6154e60baa71a2e9a1f947d326b31 249B / 249B   1.2s
 => => sha256:5435b2dcdf5cb7faa0d5b1d4d54be2c72a776 29.78MB / 29.78MB   5.0s
 => => extracting sha256:5435b2dcdf5cb7faa0d5b1d4d54be2c72a776fab9a60   1.1s
 => => extracting sha256:935d60c8480a502328adefa7698ca61261c8dc087236   0.2s
 => => extracting sha256:bed04db721b4ee2f43a7fc743a2fd7ec6181f9444d36   0.8s
 => => extracting sha256:ddf3b231f5678d6154e60baa71a2e9a1f947d326b319   0.0s
 => [internal] load build context                                       0.1s
 => => transferring context: 66B                                        0.0s
 => [2/3] WORKDIR /app                                                  0.3s
 => [3/3] COPY app.py .                                                 0.1s
 => exporting to image                                                  0.5s
 => => exporting layers                                                 0.2s
 => => exporting manifest sha256:3b17aaf93688edae6c049a904caa342c41f9   0.0s
 => => exporting config sha256:c910bd31d227d2beba5155216a5bb2af5b865e   0.0s
 => => exporting attestation manifest sha256:d163a69ecb4d3becefd5f3bf   0.1s
 => => exporting manifest list sha256:c086f0f88a73008fc6ecdbac3257374   0.0s
 => => naming to docker.io/library/docker-lab-app:latest                0.0s
 => => unpacking to docker.io/library/docker-lab-app:latest             0.1s
```

Task 2: Run Container

docker run docker-lab-app

Expected output:

Hello from Docker App

```
C:\Users\SIDDESHWARAPRASAD KR\Desktop\docker-lab>docker run docker-lab-app
Hello from Docker App
```

5. Push Image to Docker Hub

Task 1: Login to Docker Hub

docker login

```
C:\Users\SIDDESHWARAPRASAD KR\Desktop\docker-lab>docker login
Authenticating with existing credentials... [Username: siddu03]

i Info → To login with a different account, run 'docker logout' followed by
'docker login'


Login Succeeded
```

Task 2: Tag Image

docker tag docker-lab-app <your-username>/docker-lab-app

```
C:\Users\SIDDESHWARAPRASAD KR\Desktop\docker-lab>docker tag docker-lab-app s
iddu03/docker-lab-app
```

Task 3: Push Image

docker push <your-username>/docker-lab-app

```
C:\Users\SIDDESHWARAPRASAD KR\Desktop\docker-lab>docker tag docker-lab-app s
iddu03/docker-lab-app

C:\Users\SIDDESHWARAPRASAD KR\Desktop\docker-lab>docker push siddu03/docker-
lab-app
Using default tag: latest
The push refers to repository [docker.io/siddu03/docker-lab-app]
5435b2dcdf5c: Pushed
ddf3b231f567: Pushed
dd08035886f5: Pushed
bed04db721b4: Pushed
c3776cc9e0e5: Pushed
e3eb0896a3ef: Pushed
935d60c8480a: Pushed
latest: digest: sha256:c086f0f88a73008fc6ecdbac32573744800c81fa64d2957ba3a6f
57ca6e249a0 size: 856
```

Task 4: Verify

- Go to Docker Hub
- Check if image is uploaded

Name	Tag	Image ID	Created	Size	Actions
hello-world	latest	f9078146db2e	25 days ago	25.87 KB	
ubuntu	latest	c4a8d5503dfb	7 days ago	119.27 MB	
Siddu03/docker-lab-a	latest	c086f0f88a73	8 minutes agc	182.87 MB	
docker-lab-app	latest	c086f0f88a73	8 minutes agc	182.87 MB	
siddu03/docker-lab-a	latest	c086f0f88a73	8 minutes agc	182.87 MB	